导出系统

1.导出文件系统

2.制作rootfs镜像

3.解包

4.打包

1.导出文件系统

1、RK3588上的Ubuntu系统需要安装：rsync；
2、PC电脑（虚拟机）ubuntu上安装：ssh、rsync；

3.3588 开发板 ssh登录，用户root 密码fanconn123

4.先在PC主机（自己电脑为Ubuntu）上创建一个目录，用来同步保存开发板上的文件系统：

mkdir Ubuntu

sudo rsync -avx root@192.168.31.80:/ **./Ubuntu**

192.168.31.80 ： 3588IP地址

Ubuntu ： 存放的文件夹

2.制作rootfs镜像

ubuntu.img : rootfs镜像文件

ubuntu-mount ： 挂载目录

ubuntu_rootfs/* : 从板卡拷贝的文件夹

```bash
dd if=/dev/zero of=ubuntu.img bs=1M count=6144
sudo mkfs.ext4 -F -L linuxroot ubuntu.img
mkdir ubuntu-mount
sudo mount ubuntu.img ubuntu-mount
sudo cp -rfp ubuntu_rootfs/* ubuntu-mount
sudo umount ubuntu-mount
e2fsck -p -f ubuntu.img
resize2fs -M ubuntu.img
```

3.解包

1.将解压后的固件update.img ,固件名称不是update.img 的需要命名为update.img

2.将update.img 复制到 rockdev 下

3.执行 ./unpack.sh， 解包后文件在out 目录下

```
start to unpack update.img ...
********rkImageMaker ver 2.23********
Unpacking image, please wait ...
Exporting boot.bin
Exporting firmware.img
Unpacking image success.
Android Firmware Package Tool v2.2
Check file ... OK
------- UNPACK ------
package-file    offset=0x800    size=0xCB
parameter.txt   offset=0x1000   size=0x1FB
MiniLoaderAll.bin       offset=0x1800   size=0x731C0
uboot.img       offset=0x75000  size=0x400000
misc.img        offset=0x475000 size=0xC000
boot.img        offset=0x481000 size=0x2519600
recovery.img    offset=0x299A800        size=0x2DC4800
oem.img offset=0x575F000        size=0x10A6000
rootfs.img      offset=0x6805000        size=0x199B65000
Unpack firmware OK!
------ OK ------
Unpacking update.img OK.
Press any key to quit:
```

解包完成后

output目录

..	
Image	
MiniLoaderAll.bin	460
package-file	1
parameter.txt	1

image目录

..	
boot.img	37 989
misc.img	48
oem.img	17 048
recovery.img	46 866
rootfs.img	6 712 724
uboot.img	4 096

4.打包

1.将所需文件替换 output 文件夹下文件

1.1将rk3588-mkupdate.sh 复制到output目录，然后修改脚本

```
#!/bin/bash
pause()
{
echo "Press any key to quit:"
read -n1 -s key
exit 1
}
echo "start to make update.img..."
if [ ! -f "parameter.txt" -a ! -f "Image/parameter.txt" ]; then
  echo "Error:No found parameter!"
  exit 1
fi
if [ ! -f "package-file" ]; then
  echo "Error:No found package-file!"
  exit 1
fi
../afptool -pack ./ Image/update.img || pause
../rkImageMaker -RK3588 Image/MiniLoaderAll.bin Image/update.img update.img -os_type:androidos || pause
echo "Making ./Image/update.img OK."
#echo "Press any key to quit:"
#read -n1 -s key
exit $?
```

1.2 将Image下文件复制到output目录

1.3 将output 下 MiniLoaderAll.bin

package-file

parameter.txt 复制到Image目录

2.执行 ./rk3588-mkupdate.sh 进行二次打包

output目录下会生成update.img

```
start to make update.img ...
Android Firmware Package Tool v2.2
------ PACKAGE ------
Add file: ./package-file
package-file,Add file: ./package-file done,offset=0x800,size=0xcb,userspace=0x1
Add file: ./parameter.txt
parameter,Add file: ./parameter.txt done,offset=0x1000,size=0x207,userspace=0x1,flash_address=0x00000000
Add file: ./MiniLoaderAll.bin
bootloader,Add file: ./MiniLoaderAll.bin done,offset=0x1800,size=0x731c0,userspace=0xe7
Add file: ./uboot.img
uboot,Add file: ./uboot.img done,offset=0x75000,size=0x400000,userspace=0x800,flash_address=0x00004000
Add file: ./misc.img
misc,Add file: ./misc.img done,offset=0x475000,size=0xc000,userspace=0x18,flash_address=0x00006000
Add file: ./boot.img
boot,Add file: ./boot.img done,offset=0x481000,size=0x2519600,userspace=0x4a33,flash_address=0x00008000
Add file: ./recovery.img
recovery,Add file: ./recovery.img done,offset=0x299a800,size=0x2dc4800,userspace=0x5b89,flash_address=0x00028000
Add file: ./oem.img
oem,Add file: ./oem.img done,offset=0x575f000,size=0x10a6000,userspace=0x214c,flash_address=0x00078000
Add file: ./rootfs.img
rootfs,Add file: ./rootfs.img done,offset=0x6805000,size=0x199b65000,userspace=0x3336ca,flash_address=0x000b8000
Add CRC ...
Make firmware OK!
------ OK ------
********rkImageMaker ver 2.23********
Generating new image, please wait ...
Writing head info ...
Writing boot file ...
Writing firmware ...
Generating MD5 data ...
MD5 data generated successfully!
New image generated successfully!
Making ./Image/update.img OK.
```